

# INFORMATICS PRACTICES



11149

TEXTBOOK FOR CLASS XI



विद्यया ऽ मृतमश्नुते



एन सी ई आर टी  
NCERT

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## FOREWORD

Information Technology has continuously been crossing the barriers of access and communication and reaching more and more people. The number of internet users in India has been on the rise. The tremendous growth in computer science, telecommunications and information technology has resulted in automation of various tasks and contributed to the ease of living. Technology has made continuous inroads into diverse areas—be it business, commerce, science, sports, health, transportation or education. Today, we are living in an interconnected world where computer based applications influence the way we learn, communicate, commute, or even socialise.

With so many users of information and communication technology (ICT), huge volumes of data are continuously generated at an unprecedented rate. Many innovative business models are being evolved which utilise such data to reach potential customers in a more targeted way. Government agencies are also using data to deliver services and fast track progress of different programmes, strengthen accountability and to make more informed decisions. This has been creating better opportunities for our youth not only to enter the field of technical education but also in the world of work. NCERT, for the first time, has developed a textbook on 'Informative Practices' to develop skill sets in students to make use of the opportunities provided by ICT.

This book focuses on the fundamental concepts related to handling of data while opening a window to the emerging areas of data processing. It seeks to address the dual challenges of reducing curricular load as well as introducing the latest development in the field of ICT.

As an organisation committed to systemic reforms and continuous improvement in the quality of its curricular material, NCERT welcomes comments and suggestions to enable us to bring about necessary changes in its further publications.

New Delhi  
July 2019

HRUSHIKESH SENAPATY  
*Director*  
National Council of Educational  
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# THE CONSTITUTION OF INDIA

## PREAMBLE

**WE, THE PEOPLE OF INDIA**, having solemnly resolved to constitute India into a <sup>1</sup>**[SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC]** and to secure to all its citizens :

**JUSTICE**, social, economic and political;

**LIBERTY** of thought, expression, belief, faith and worship;

**EQUALITY** of status and of opportunity; and to promote among them all

**FRATERNITY** assuring the dignity of the individual and the <sup>2</sup>[unity and integrity of the Nation];

**IN OUR CONSTITUENT ASSEMBLY** this twenty-sixth day of November, 1949 do **HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.**

1. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Sovereign Democratic Republic" (w.e.f. 3.1.1977)
2. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Unity of the Nation" (w.e.f. 3.1.1977)

## PREFACE

In the present education system of our country, specialised/discipline based courses are introduced at the higher secondary stage. This stage is crucial as well as challenging because of the transition from general to discipline-based curriculum. The syllabus at this stage needs to have sufficient rigour and depth while remaining mindful of the comprehension level of the learners. Further, the textbook should not be heavily loaded with content.

We are living in an era where information drives many of our socio-economic decisions. Millions of people are accessing internet round the clock for availing various services and thereby generating vast amount of data. Processing of data is becoming a key skill with applications across the disciplines. Thus, study of basic concepts of data handling and analysis is becoming more and more desirable. There are courses offered in the name of computer science, Information and Communication Technology (ICT), Information Technology (IT), etc. by various boards and schools up to the secondary stage, as an optional subject. These mainly focus on using computer for word processing, presentation tools and application software.

Informatics Practices (IP) at the higher secondary stage of school education is also offered as an optional subject. At this stage, students can take up IP with the aim of pursuing a career in data science or related areas after going through professional courses at higher levels. Therefore, at the higher secondary stage, the curriculum of IP introduces the basics of database management systems and data processing. The book has eight chapters covering the following broader themes:

- Basic understanding of computer systems and their evolution, introduction to software and their categorisation, computer memory, awareness of emerging trends in the field of information and communication technology.
- Basic constructs of a program using Python programming language — program structure, identifiers, variables, flow of control, advanced data types like Lists and Dictionaries.
- Handling data using specialised Python library called NumPy — concept of single and multi-dimensional Array.
- Concepts of data, database, and relational database management system using MySQL. Structured query language — data definition, data manipulation and data querying.

Python programming language and NumPy are introduced using both the interactive and script mode. A number of hands-on examples are given in Python, NumPy and MySQL to gradually explain the methodology to solve different types of problems and handle data. The programming and database related examples as well as the exercises in those chapters are required to be solved in a computer and verified with the given outputs.

The chapters in this book have two additional components — activities for self assessment and ‘think and reflect’ to generate further interest in the learner.

Group projects through case studies are proposed to solve complex problems. Some exercises have been made in case-study form to promote problem-finding and problem-solving skills.

These chapters have been written by involving practicing teachers as well as subject experts. These have been iteratively peer-reviewed. Several iterations have resulted into this book. Thanks to the authors and reviewers for their valuable contribution.

Comments and suggestions are welcome to make this endeavour par excellence.

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